Search-Powered AI Applications: Engineering Beyond the Demo

What it actually takes to build production multimodal AI systems

Elastic Chennai Meetup

TALK

Search-Powered AI Applications: Engineering Beyond the Demo

Jeyajeevanram S

Senior Engineer @ SQ1

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10:00 am - 2:00 pm

SQ1 Sholinganallur

elastic

The AI Industry Right Now

What people talk about

- Prompt engineering
- Model benchmarks
- "Which LLM is best?"
- Viral demos

What production teams actually deals with

- Retrieval failures
- Latency spikes
- GPU bottlenecks
- Broken pipelines
- Context quality issues
- Operational reliability

THE DEMO GOT APPLAUSE.

THE PRODUCTION DEPLOYMENT GOT ALERTS.

Demos vs. Production

Demo	Production
• Best-case data	• Real, noisy data
• Single user	• Massive concurrency
• Manual fixes allowed	• Full automation required
• Short-lived	• Operationally reliable
• Optimized for excitement	• Optimized for **reliability**

What Were We Actually Trying to Solve?

Users wanted to:

- Search with screenshots
- Query PDFs
- Understand diagrams
- Search logs contextually
- Identify patterns
- Ask natural-language questions

The core challenge:

"How do we make unstructured multimodal enterprise data searchable?"

The Naive RAG Pipeline

Document
Collect raw source files

Embed
Convert text to vectors

Chunk
Split into smaller passages

Retrieve
Search for relevant vectors

What went wrong

- Irrelevant retrieval, weak chunking
- Hallucinations from bad context
- Token explosion, poor precision
- Latency spikes at every step

⚠ **Key realization**

"This wasn't just an LLM problem. It was a **search problem.**"

Keyword Search Was Not Enough

Ingestion inconsistencies

Noisy extraction corrupted search indexes

Semantic mismatch

Keyword matching missed conceptual meaning

Context fragmentation

Documents split across disconnected chunks

Image-heavy content

Visual data completely invisible to search

Missing metadata

No filtering, no structure, no precision

Pure Vector Search Was Not Enough

Wrong results
Semantically similar but factually incorrect

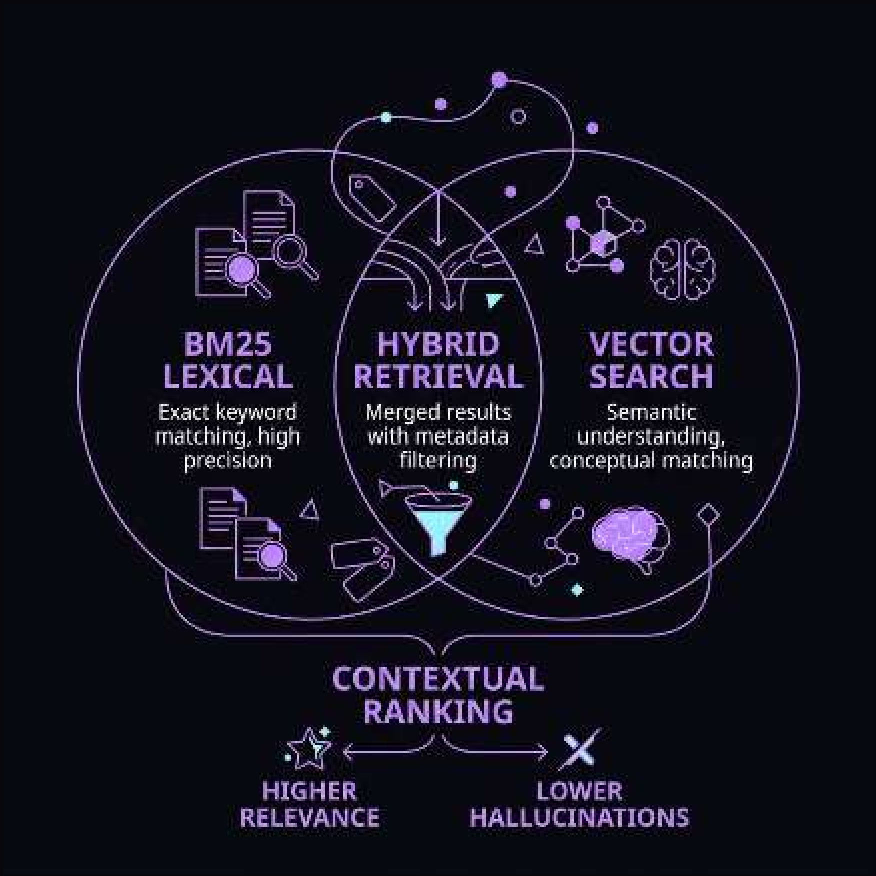
Weak precision
Fuzzy matches with no lexical grounding

Metadata blindness
No structural filtering at query time

Poor explainability
Opaque scoring, hard to debug failures

Production AI depends on finding the **right context** — quickly, accurately, at scale.

BM25 + Vector Search Changed Everything



What the combination unlocked

- Higher relevance scores
- Fewer context drift
- Better retrieval quality
- Fewer useless tokens sent to inference

"Better retrieval improved AI quality more than changing models."

What critical problems elastic solved

Elastic gave us:

Hybrid search

BM25 + vector retrieval combined

Metadata filtering

Precision at query time

Distributed scalability

Realtime indexing at enterprise scale

Operational maturity

Battle-tested reliability

Metadata Improved Precision Dramatically

Document Types

Classify content at ingest for targeted filtering

Timestamps

Time-bound retrieval eliminates stale results

Source Systems

Provenance filtering improves trust and precision

Operational Tags

User context + categories enable scoped queries

"Structured filtering + semantic retrieval is incredibly powerful."

Chunking Changed Everything

What actually worked

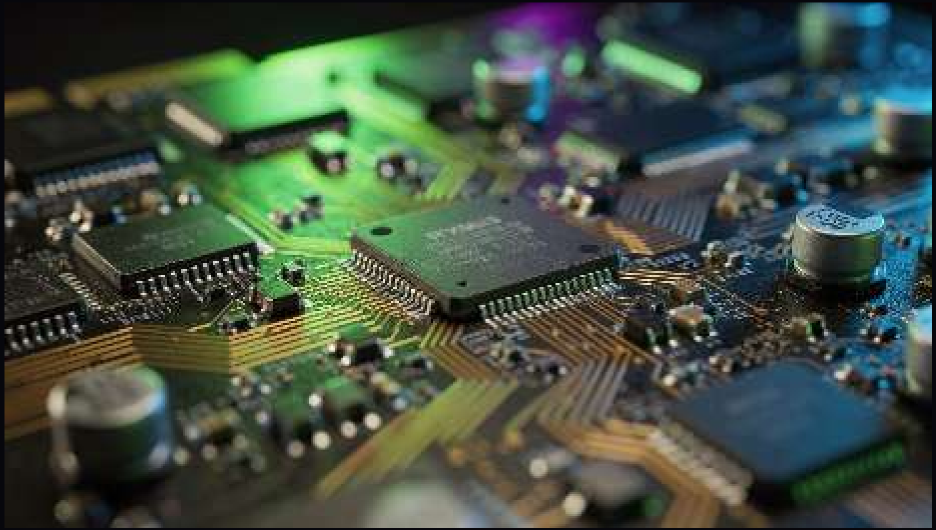
- Semantic chunking by meaning boundaries
- Structural awareness of document layout
- Metadata enrichment at chunk time

"Bad chunking creates retrieval drift."

AI Is Becoming Retrieval-Centric

- Hybrid search
- Multimodal indexing
- Realtime retrieval
- Contextual memory
- Scalable inference

Efficient Inference Actually Matters



vLLM improved:

Throughput

More requests per GPU

Concurrency

Parallel request handling

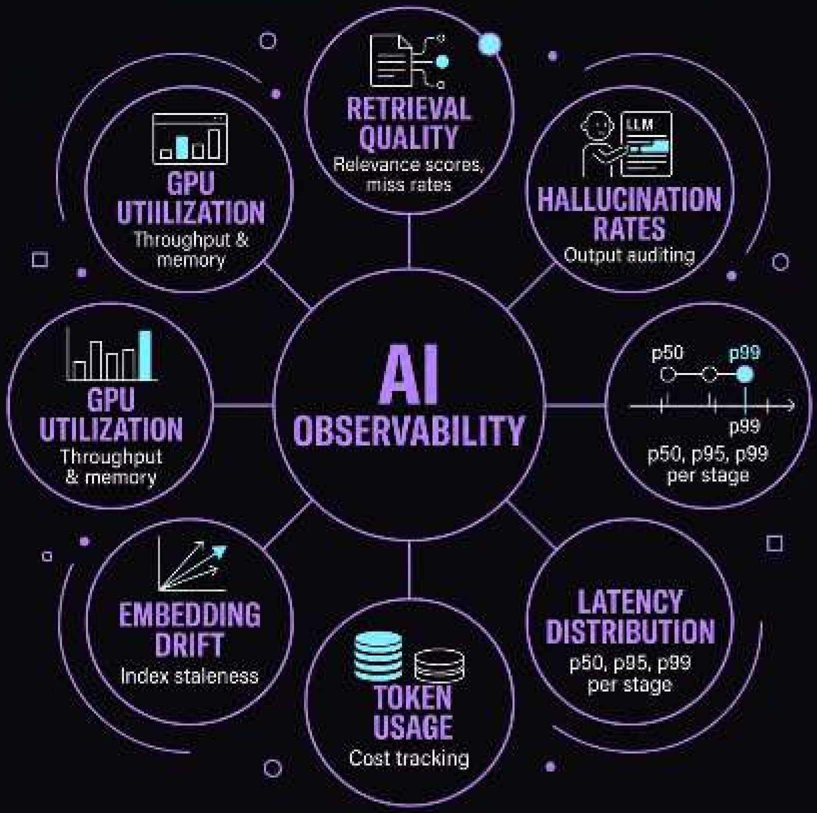
Memory Efficiency

PagedAttention reduces waste

Token Streaming

Perceived latency drops significantly

AI Systems Need Deep Observability



Traditional APM monitoring misses the AI-specific failure modes. You need visibility into every stage of the retrieval pipeline.

⚠ **"You cannot debug invisible retrieval failures."**

Real Production Failures

Wrong context retrieval

Semantically plausible but factually wrong chunks returned

Stale embeddings

Index drift after document updates caused silent degradation

Retrieval amplification

Bad queries triggered cascading over-retrieval

Ingestion corruption

Garbled extraction poisoned downstream retrieval permanently

Token overflows

Over-retrieved context exceeded model context windows

Why Elastic Was Critical in Production

Elastic solved the hard operational problems

01

Scalable indexing

Ingestion at enterprise volume

02

Distributed retrieval

Low-latency search across shards

03

Metadata-aware filtering

Precision queries at scale

04

Operational resilience

Built-in replication and recovery

Why Elasticsearch Worked Better in Production AI Systems

Elasticsearch

- Hybrid Search (BM25 + Vector)
- Advanced Metadata Filtering
- Realtime Indexing
- Explainable Retrieval
- Enterprise-Scale Distributed Architecture
- Unified Retrieval Platform
- Better Context Precision
- Reduced Token & GPU Costs

Pure Vector Databases

- Semantic Search Only
- Limited Filtering
- Slower Update Pipelines
- Opaque Similarity Scoring
- Vector-Focused Scaling
- Multiple Supporting Systems Required
- More Irrelevant Retrieval
- Higher Retrieval Inefficiency

Platform Consolidation & Simplified Architecture

Typical RAG Systems

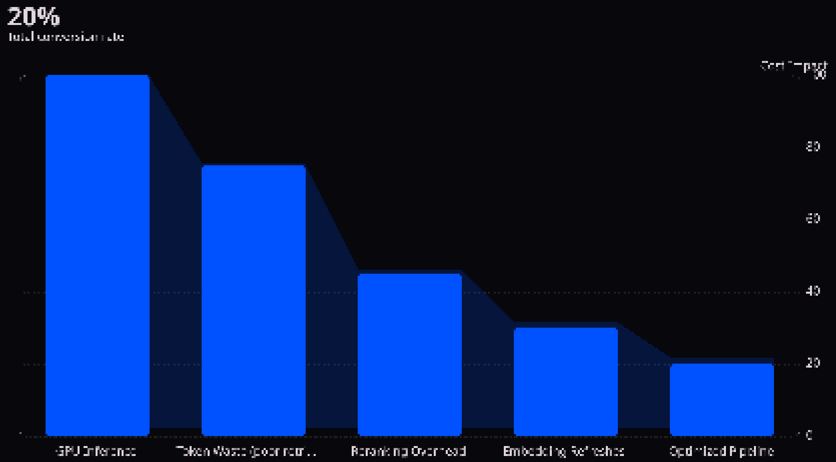
- Vector DB
- Keyword Search Engine
- Analytics System
- Observability Stack
- Reranking Service
- Orchestration Glue

CONSOLIDATE →

Elastic Platform

- Hybrid Search
- Analytics
- Reranking
- Observability
- Filtering

The Economics of AI



Better retrieval reduces:

- Contextual drift and inference instability
- Token inefficiency from low-signal context
- Per-query inference and compute overhead

Retrieval optimization is your best cost lever — before buying more GPUs.

Questions?

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